

Mutual Fund Analytics Platform

An End-to-End Financial Data Analytics & Visualization System

Data Analytics Capstone Project

Project Type

Data Analytics Capstone Project — Investment Performance & Risk Analytics

Python

Pandas

SQLite

Power BI

Jupyter Notebook

Technologies Used

Prepared For

Bluestock Fintech — Mutual Fund Analytics Initiative

Domain

Financial Data Analytics / Investment Research

Tools & Platform

Python, Pandas, SQLite, Power BI, Jupyter Notebook

Report Type

Final Capstone Project Submission

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1. Executive Summary

The mutual fund industry generates a continuous stream of Net Asset Value (NAV) records, transaction data, and scheme-level information that is difficult for individual investors and analysts to interpret without structured tooling. Investors are frequently required to compare hundreds of schemes across return, volatility, and cost dimensions, yet raw fund data is scattered, inconsistently formatted, and rarely analysis-ready. The **Mutual Fund Analytics Platform** addresses this gap by building a complete, automated analytics pipeline that ingests raw mutual fund data, cleans and structures it, calculates industry-standard performance and risk metrics, and delivers the results through an interactive Power BI dashboard.

Mutual fund analytics is important because investment decisions are ultimately risk-return trade-offs. A fund with a high headline return may carry disproportionate volatility, while a fund with a modest return may offer superior risk-adjusted performance. Without consistent metrics such as CAGR, Sharpe Ratio, Alpha, and Beta, investors cannot meaningfully compare schemes across fund houses and categories. This project brings that comparability to the data by standardizing calculations across all schemes in the dataset.

The resulting system provides two layers of value. First, it gives investors and analysts a performance layer — daily returns, CAGR over multiple horizons, and fund-versus-benchmark comparison. Second, it provides a risk layer — volatility, Sharpe Ratio, Beta, and Value at Risk (VaR) — that quantifies the downside an investor is exposed to for a given return. Together, these layers are surfaced through a four-page Power BI dashboard that allows any user to explore the mutual fund universe, drill into individual schemes, and identify funds that match their risk appetite.

Key Outcomes

- *A reproducible ETL pipeline that converts raw CSV exports into an analysis-ready SQLite database.*
- *A standardized performance layer covering daily returns and 1/3/5-year CAGR.*
- *A risk layer covering volatility, Sharpe Ratio, Alpha, Beta, and Value at Risk.*
- *An interactive four-page Power BI dashboard for industry overview, performance, risk, and fund exploration.*

2. Project Objectives

The project was scoped around seven core objectives, spanning data engineering, analytics, and visualization. Each objective builds on the previous stage, forming a complete pipeline from raw data to decision-ready insight.

#	Objective	Description
1	Automated ETL Pipeline	Build a repeatable pipeline that ingests, cleans, and loads mutual fund data without manual intervention.
2	Data Cleaning & Processing	Standardize and validate raw mutual fund datasets, handling missing values, duplicates, and type mismatches.
3	SQLite Storage	Persist processed data into a structured relational database for efficient querying and downstream analysis.
4	Exploratory Data Analysis	Understand NAV distributions, data quality, and fund-level trends before performance modelling.
5	Performance Metrics	Calculate daily returns, CAGR, Sharpe Ratio, Alpha, and Beta for each scheme.
6	Interactive Power BI Dashboard	Visualize industry, performance, and risk insights through a multi-page dashboard.
7	Advanced Analytics Models	Extend the platform with Value at Risk, a fund recommendation engine, and volatility-based risk analysis.

3. Dataset Description

The platform works with two categories of data: raw source files that mirror how mutual fund information is typically obtained (scheme master data, historical NAVs, transactions, and live NAV feeds), and processed files that represent the cleaned, analysis-ready output of the ETL pipeline.

3.1 Raw Data

File	Description
fund_master.csv	Scheme-level master data — scheme code, fund name, category, fund house, and expense ratio.
nav_history.csv	Historical daily NAV records for every scheme, used to compute returns and trends.
transactions.csv	Investor-level transaction records including purchases, redemptions, and switches.
live_nav.csv	Latest available NAV snapshot used to refresh dashboards with near-real-time values.

3.2 Processed Data

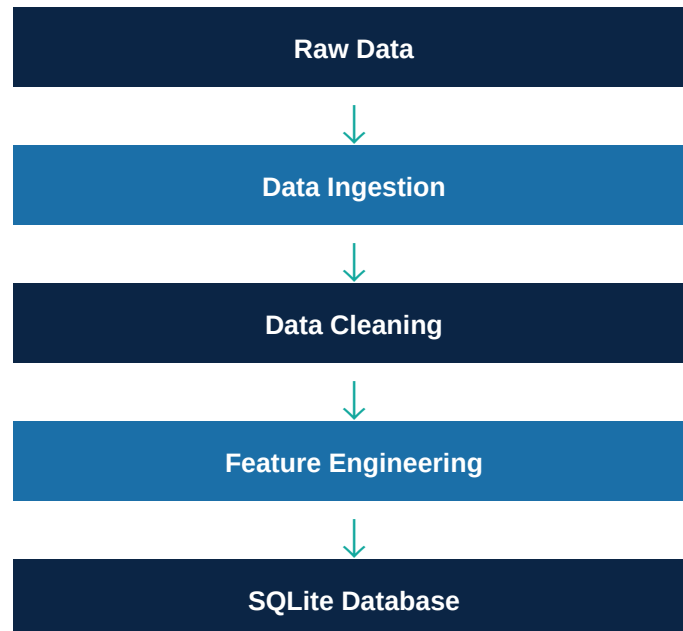
File	Description
clean_nav_history.csv	De-duplicated, type-validated NAV history with missing values handled and dates normalized.
clean_performance.csv	Derived performance metrics per scheme — daily returns, CAGR, Sharpe Ratio, Alpha, Beta.
clean_transactions.csv	Validated transaction records with standardized transaction types and reconciled amounts.

3.3 Key Columns

Column	Description
scheme_code	Unique identifier for a mutual fund scheme; the primary key used to join all datasets.
NAV	Net Asset Value — the per-unit price of the fund on a given date; the basis for all return calculations.
date	Trading date associated with a NAV or transaction record, used for time-series analysis.
transaction_type	Classifies a transaction as purchase, redemption, switch-in, or switch-out.
expense_ratio	Annual fee charged by the fund house as a percentage of assets, used in risk-adjusted comparisons.

4. ETL Pipeline

The ETL (Extract, Transform, Load) pipeline is the backbone of the platform. It converts inconsistent raw CSV exports into a validated, query-ready SQLite database through four sequential stages, each implemented as an independent, testable Python module.



4.1 Implementation Approach

- **Python scripts** — each pipeline stage is a standalone script that can be run independently or orchestrated end-to-end.
- **pathlib-based file handling** — all file paths are resolved using Python's `pathlib` module for cross-platform reliability.
- **Error handling** — ingestion and cleaning steps wrap file reads and transformations in structured exception handling with logging, preventing partial or silent failures.
- **Automated execution** — the full pipeline is triggered through a single orchestrator script, making the process repeatable for scheduled or ad-hoc runs.

4.2 Source Structure

```
src/
  ■■■ data_ingestion.py    # Reads raw CSVs, validates schema
  ■■■ data_cleaning.py     # Handles nulls, duplicates, type casting
  ■■■ load_database.py     # Writes processed data into SQLite
  ■■■ etl_pipeline.py     # Orchestrates the full ETL sequence
```

5. Database Design

Processed data is persisted in a single SQLite database, `bluestock_mf.db`, structured around three fact tables that together support performance and risk queries across the full mutual fund universe.

5.1 fact_nav

Column	Description
scheme_code	Foreign key referencing the fund scheme.
date	Trading date of the recorded NAV.
nav	Net Asset Value on that date.

5.2 fact_transactions

Column	Description
transaction_id	Unique identifier for each transaction record.
scheme_code	Scheme associated with the transaction.
transaction_type	Purchase, redemption, switch-in, or switch-out.
amount	Transaction value in Indian Rupees.
date	Date the transaction was executed.

5.3 fact_performance

Column	Description
scheme_code	Scheme the performance record belongs to.
expense_ratio	Annual expense ratio used in cost and risk-adjusted comparisons.

5.4 Why SQLite

SQLite was selected because the platform needs a lightweight, file-based relational database rather than a client-server system. It requires no separate installation or network configuration, integrates natively with Python through the standard `sqlite3` module, and connects directly to Power BI as a data source. For a capstone-scale dataset of scheme, NAV, and transaction records, SQLite offers sufficient performance while keeping the entire database portable as a single `.db` file — well suited for reproducible analytics work and easy handoff between development and reporting environments.

6. Exploratory Data Analysis

Before performance metrics were calculated, exploratory data analysis (EDA) was carried out to understand the shape, quality, and behavior of the cleaned datasets. This step ensures that downstream performance and risk calculations are built on a reliable foundation.

- **NAV distribution analysis** — examined the spread and skew of NAV values across schemes and categories.
- **Missing value checks** — quantified null and gap patterns in NAV history and transaction records prior to cleaning.
- **Fund comparisons** — compared NAV trajectories and category-level behavior across equity, debt, and hybrid schemes.
- **Performance trends** — reviewed NAV movement over time to identify seasonality, volatility clusters, and outliers.

■ *Insert EDA charts here — NAV distribution, missing value heatmap, fund comparison plots*

EDA Insights

- *NAV movement patterns show that equity-oriented schemes exhibit higher day-to-day fluctuation than debt schemes, consistent with their higher-risk mandates.*
- *Fund performance differences are most pronounced during volatile market periods, where category and fund-house selection materially affects returns.*
- *Data quality observations indicate that missing NAV values are concentrated around non-trading days and were resolved through forward-fill logic during cleaning, rather than indicating systemic data issues.*

7. Performance Analytics

Performance analytics translate raw NAV history into metrics that are directly comparable across schemes. The platform calculates four categories of performance and risk-adjusted measures.

7.1 Daily Returns

Daily returns measure the percentage change in NAV from one trading day to the next, forming the base series from which volatility, Sharpe Ratio, and other metrics are derived.

Daily Return Formula

$$\text{Return} = (\text{NAV}_{\text{today}} / \text{NAV}_{\text{previous day}}) - 1$$

7.2 CAGR (Compound Annual Growth Rate)

CAGR expresses the smoothed annual growth rate of an investment over a defined holding period, removing the noise of day-to-day fluctuations. The platform reports CAGR across three standard investment horizons so that funds can be compared on a like-for-like basis regardless of when an investor entered the scheme.

- **1-Year CAGR** — short-term growth rate, useful for gauging recent momentum.
- **3-Year CAGR** — medium-term growth rate that smooths out short-term market cycles.
- **5-Year CAGR** — long-term growth rate, the most reliable indicator of a fund's consistency.

7.3 Sharpe Ratio

The Sharpe Ratio measures risk-adjusted return — how much excess return a fund generates per unit of volatility it takes on, relative to a risk-free rate. Two funds with identical returns can have very different Sharpe Ratios if one achieves that return with significantly more volatility. A higher Sharpe Ratio indicates more efficient use of risk.

7.4 Alpha and Beta

Alpha and Beta position a fund's performance relative to a market benchmark. **Beta** measures a fund's sensitivity to benchmark movements — a Beta above 1 indicates the fund is more volatile than the market, while a Beta below 1 indicates lower relative volatility. **Alpha** measures the excess return a fund generates beyond what its Beta would predict, effectively capturing the value added by active fund management after adjusting for market risk.

■ *Insert Performance charts here — CAGR comparison, Sharpe Ratio ranking, return trend lines*

8. Power BI Dashboard

The processed SQLite database feeds directly into a four-page Power BI dashboard, giving users a progression from a high-level industry view down to individual scheme detail.

Page 1 — Industry Overview

- Fund count across categories and fund houses
- Performance summary at an industry level
- NAV trends over time

Page 2 — Performance Analysis

- CAGR comparison across schemes
- Fund ranking by return metrics

Page 3 — Risk Analysis

- Sharpe Ratio by scheme
- Beta relative to benchmark
- Risk comparison across categories

Page 4 — Fund Explorer

- Detailed fund information
- NAV history and trend view
- Performance metrics for a selected scheme

■ *Insert Dashboard screenshots — Industry Overview, Performance Analysis, Risk Analysis, Fund Explorer*

9. Advanced Analytics

9.1 Value at Risk (VaR)

Purpose: Value at Risk estimates the maximum expected loss on a fund over a defined time horizon at a given confidence level, providing investors with a concrete downside-risk figure rather than an abstract volatility measure.

9.2 Fund Recommendation System

Logic: Schemes are ranked using a performance-based scoring approach that blends return metrics (CAGR) with risk-adjusted measures (Sharpe Ratio) and cost (expense ratio), surfacing funds that balance strong performance with efficient risk management rather than simply ranking by raw return.

9.3 Risk Analysis

- **Volatility** — standard deviation of daily returns, used as the core measure of a fund's risk profile.
- **Risk comparison** — side-by-side comparison of volatility and VaR across schemes and categories to support portfolio-level decisions.

10. Results and Findings

The completed platform delivers measurable value across analysis, comparison, and process efficiency:

- The dashboard provides interactive fund analysis, allowing users to move from an industry-wide view to individual scheme detail without leaving the report.
- Performance metrics allow direct, like-for-like comparison between schemes across categories and fund houses.
- Risk metrics such as Sharpe Ratio, Beta, and VaR help identify suitable investment options aligned to an investor's risk appetite, not just their return expectations.
- The automated ETL pipeline improves repeatability, allowing the entire dataset to be refreshed and reprocessed with a single pipeline run rather than manual spreadsheet work.

11. Conclusion

The Mutual Fund Analytics Platform successfully combines ETL, database management, analytics, and visualization into a complete financial analytics solution. Starting from raw, inconsistent mutual fund data, the pipeline produces a clean, structured SQLite database, layers on standardized performance and risk metrics, and exposes the results through an interactive Power BI dashboard. The result is a system that reduces the manual effort required to evaluate mutual fund schemes while improving the consistency and rigor of that evaluation — turning raw financial data into actionable investment insight.

12. Future Improvements

While the current platform meets its capstone objectives, several extensions would increase its real-world readiness:

- **Live NAV API integration** — replace static snapshot files with a scheduled live NAV feed for near-real-time dashboard updates.
- **Scheduled ETL automation** — trigger the pipeline on a recurring schedule (e.g., daily) rather than manual execution.
- **Streamlit application** — build a lightweight, code-driven front end as a complement to the Power BI dashboard for faster iteration.
- **Portfolio optimization** — extend the scoring engine into a full portfolio construction module using mean-variance or similar optimization techniques.
- **Monte Carlo simulation** — model a distribution of future NAV outcomes to strengthen the risk analysis beyond historical VaR.